

# MICHAEL HOON YONG HAU

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## EDUCATION

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| <b>Singapore University of Technology and Design</b><br><i>Bachelor of Engineering in Engineering Systems and Design, Minor in Artificial Intelligence (AI)</i> | Sep. 2022 – May 2026<br>GPA: 4.59/5.00      |
| <b>Chalmers University of Technology</b><br><i>Spring Exchange, Machine Learning, AI, Big Data Systems</i>                                                      | Jan. 2025 – Jun. 2025<br>Gothenburg, Sweden |
| <b>University of California, Berkeley</b><br><i>Summer Exchange, Econometrics</i>                                                                               | Jun. 2023 – Aug. 2023<br>Berkeley, CA       |

## EXPERIENCE

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| <b>AI Engineer Intern</b><br><i>HTX (Home Team Science &amp; Technology Agency)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sep. 2025 – Apr. 2026<br>Singapore |
| <ul style="list-style-type: none"><li>Spearheaded development of multi-agent, multi-modal LLM system with LangGraph for multi-turn forensic reasoning in complex Singapore-specific AI-generated scams, achieving ~91% system accuracy.</li><li>Built evaluation, observability, and tracing pipelines with self-hosted LangSmith platform, deploying local open-sourced LLMs with vLLM inference engine on custom hardware for low-latency processing.</li><li>Developed deployment workflow with Docker and Docker-Compose, to orchestrate containerised microservices, ensuring a fault-tolerant and reproducible environment.</li></ul>                                                                                       |                                    |
| <b>Data Engineer Intern</b><br><i>Government Technology Agency (GovTech) Singapore</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Jun. 2025 – Sep. 2025<br>Singapore |
| <ul style="list-style-type: none"><li>Optimised inefficiencies in Redshift data warehouse (column encoding, workload management), migrating provisioned clusters to serverless architecture, reducing storage costs by ~30% and query latency by ~15%.</li><li>Spearheaded an AWS SageMaker Unified Studio landing zone (Bedrock, Lake Formation, Glue, Athena, DataZone) with Terraform IaC, centralizing data access for 50+ users and cutting analytics onboarding time by ~80%.</li><li>Productionised end-to-end ETL pipeline for clickstream analytics, ingesting 100M+ daily events via Data Firehose, Glue, S3, Redshift Spectrum, PySpark. Enabled analytics and dashboarding via SageMaker AI and Quicksight.</li></ul> |                                    |
| <b>AI Engineer Intern</b><br><i>DSO National Laboratories</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mar. 2024 – Dec. 2024<br>Singapore |
| <ul style="list-style-type: none"><li>Developed a full-stack local RAG pipeline for assisting researchers with internal workflows, reducing initial research and policy analysis by 10+ hours/week. Designed pipeline for Knowledge Graph generation from PDF files.</li><li>Designed a containerized, Multi-Agent Graph-based RAG system that answered complex multi-hop queries, improving accuracy by ~30% vs. standard RAG pipelines.</li><li>Architected end-to-end RAG pipeline (LangGraph, Ollama), with Knowledge Graph database (Neo4j, MinIO S3, PostgreSQL), and LLM-as-a-Judge evaluation, with LangSmith for tracing and observability.</li></ul>                                                                    |                                    |
| <b>Data Analyst Intern</b><br><i>Poh Tiong Choon Logistics Ltd.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Jan. 2024 – Apr. 2024<br>Singapore |
| <ul style="list-style-type: none"><li>Supply chain optimization, analyzed delivery driver datasets using Python/PowerBI to derive insights and improve scheduling of drivers, boosting driver utilization by ~20%.</li><li>Built scheduled Bash/Python ETL scripts to automate data cleaning, visualization, and analysis with PowerBI, slashing weekly report generation time from 8 hours to 1 hour.</li></ul>                                                                                                                                                                                                                                                                                                                  |                                    |

## PROJECTS

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| <b>LepakLah!</b>   <i>Python, FastAPI, MySQL, Docker, Kubernetes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Aug. 2024 – Sep. 2024 |
| <ul style="list-style-type: none"><li>Delivered a Flutter + FastAPI cloud-native mobile app for a Dell public-good hackathon, reducing senior workshop activity planning time by ~60%. 4.8/5 satisfaction, won 2nd Runner up representing University.</li><li>Led backend development, deployed LLM system and Stable Diffusion workflow via Nvidia NIM to auto-generate personalized workshops. Containerised microservices with Docker/Kubernetes on Red Hat OpenShift, able to scale to support 500+ concurrent users.</li></ul>                                                                                 |                       |
| <b>Cloud-Based MRP System</b>   <i>Python, Azure, Airflow, Docker</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | May. 2024 – Aug. 2024 |
| <ul style="list-style-type: none"><li>Designed backend system for Materials Requirement Planning (MRP) for a supply chain project, optimizing inventory management by forecasting material requirements and automating workflows.</li><li>Built an end-to-end ETL pipeline with Python, SQL, Azure Cloud Platform, and orchestrated automated DAG workflows with Apache Airflow. Containerised back-end microservices with Docker.</li><li>Integrated system with Azure SQL database to store and retrieve data on Microsoft Azure Cloud Platform, enabling scalable and secure cloud-based data storage.</li></ul> |                       |

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| <b>Kaggle Transportation Analytics</b>   <i>R, Machine Learning, XGBoost</i>                                                                                                                                                                                                                                                                                                                                                                                                                    | Aug. 2024 – Aug. 2024 |
| <ul style="list-style-type: none"> <li>Engineered a stacked ensemble (Mixed Logit + XGBoost) in <i>R</i>, predicting vehicle feature preferences with ~88% accuracy, outperforming 30+ teams and winning 1st Runner up on internal University hackathon.</li> <li>Fine-tuned model hyperparameters and validated results on private test set with cross-entropy loss. Leveraged SHAP values for interpretability and identified key safety features driving user purchase decisions.</li> </ul> |                       |

## SKILLS

**Languages:** Python, SQL, R, Go  
**Tools:** Git/GitLab, AWS Cloud, Linux/Bash, Docker, Kubernetes, Terraform, Airflow, PostgreSQL, Neo4j, Kafka  
**Frameworks:** Pandas, NumPy, PySpark, LangGraph, HuggingFace, scikit-learn, FastAPI, tidyverse (R), Scrum

## AWARDS

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| <b>SUTD Honours and Research Programme</b>                                                                                                                                                                         | Sep. 2022 - May. 2026 |
| <i>Scholarship</i>                                                                                                                                                                                                 | <i>Singapore</i>      |
| <ul style="list-style-type: none"> <li>Selected as one of three students out of a cohort of 500+ for the highly competitive SUTD Honours and Research Program (SHARP), securing a \$15K research grant.</li> </ul> |                       |
| <b>SUTD Global Merit Award</b>                                                                                                                                                                                     | Sep. 2022 - May. 2026 |
| <i>Scholarship</i>                                                                                                                                                                                                 | <i>Singapore</i>      |
| <ul style="list-style-type: none"> <li>Awarded full sponsorship (worth over \$16K) under SUTD Global Leadership Programme for academic summer exchange at University of California, Berkeley.</li> </ul>           |                       |
| <b>A C Toh - SUTD Scholarship</b>                                                                                                                                                                                  | Sep. 2022 - May. 2026 |
| <i>Scholarship</i>                                                                                                                                                                                                 | <i>Singapore</i>      |
| <ul style="list-style-type: none"> <li>Awarded full sponsorship covering tuition fees, stipend, and on-campus housing (worth over \$60K) for outstanding academic achievement.</li> </ul>                          |                       |
| <b>KKH - SUTD Exchange Award</b>                                                                                                                                                                                   | Jan. 2025 - Jun. 2025 |
| <i>Award</i>                                                                                                                                                                                                       | <i>Singapore</i>      |
| <ul style="list-style-type: none"> <li>Awarded sponsorship worth \$5K to support overseas semester exchange in Sweden.</li> </ul>                                                                                  |                       |
| <b>IMDA SG Digital Young Leaders Programme</b>                                                                                                                                                                     | Sep. 2024 - May. 2025 |
| <i>Talent Programme</i>                                                                                                                                                                                            | <i>Singapore</i>      |
| <ul style="list-style-type: none"> <li>Selected as part of IMDA's talent development initiative for mentorship and networking through industry attachments.</li> </ul>                                             |                       |

## LEADERSHIP

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| <b>Secretary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Mar. 2023 – Feb. 2024 |
| <i>SUTD Organisation of Autonomous Robotics (SOAR)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Singapore</i>      |
| <ul style="list-style-type: none"> <li>Organised a university-wide robotics competition (RoboClash 2024), attracting over 100+ participants across 20+ teams, and judging coordination to ensure a successful large-scale event.</li> <li>Spearheaded sponsorship and partnership initiatives, successfully securing \$50K+ in funding from external companies and university grants to support competitions, equipment upgrades, and outreach programs.</li> <li>Managed club operations, communications, and documentation, ensuring smooth coordination across project teams and executive committee members.</li> </ul> |                       |